
SOFTWARE REQUIREMENTS SPECIFICATION

Hotel Management System

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Hotel Management System

Software Requirements Specification

1. Abstract ---

The Hotel Management System is a software application designed to automate and streamline the day-to-day operations of hotels, including room reservations, guest management, check-in and check-out processes, billing, staff management, and report generation. The system provides a centralized platform that helps hotel administrators efficiently manage resources, improve customer service, and maintain accurate records while reducing manual effort and operational errors.

This project demonstrates the application of software engineering principles in developing a comprehensive management solution, showcasing key concepts such as modular design, database management, object-oriented programming, and user-friendly interfaces. It enables hotel staff to perform various operational tasks efficiently, enhances the guest experience through faster service delivery, and provides management with real-time access to important business information, all within a structured and integrated system.

2. Existing System ---

Before the introduction of computerized Hotel Management Systems, hotel operations were managed manually using paper records, registers, spreadsheets, or disconnected software applications. These traditional systems lacked automation, centralized data management, and real-time access to information, resulting in inefficiencies and operational challenges.

The problems found in existing systems include:

2.1 Manual Reservation Management

Traditional hotel management relied heavily on manual booking processes, leading to:

- Errors in room reservations and guest records
- Double booking or overbooking of rooms
- Difficulty in tracking room availability in real time

2.2 Inefficient Guest Management

Older systems lacked centralized guest information management, causing:

- Time-consuming guest registration processes
- Difficulty maintaining guest history and preferences
- Delays in providing personalized customer service

2.3 Limited Billing and Payment Processing

Existing systems often handled billing manually, resulting in:

- Calculation errors in invoices and payments
- Delays in generating bills during check-out
- Difficulty tracking payment status and transaction history

2.4 Lack of Integrated Operations

Traditional hotel management methods operated as separate processes without integration, leading to:

- Poor coordination between reception, housekeeping, and management
- Delayed updates on room status and occupancy
- Increased administrative workload and communication gaps

2.5 Limited Reporting and Analytics

Earlier systems provided minimal reporting capabilities, causing:

- Difficulty monitoring hotel performance and occupancy rates
- Limited access to business insights and financial reports
- Challenges in making data-driven management decisions

3. Proposed System

The proposed Hotel Management System automates the entire hotel operation process through a centralized software platform. It streamlines room reservations, guest management, billing, staff management, and reporting activities, enabling hotels to operate efficiently while providing enhanced customer service.

3.1 System Provides

- Automated room reservation and booking management
- Real-time room availability tracking
- Guest registration and profile management
- Check-in and check-out processing
- Staff information and task management
- Report generation and business analytics
- Secure data storage and retrieval

3.2 Key Improvements Over Existing Systems

- Centralized management of all hotel operations
- Real-time room availability and booking updates
- Faster guest registration and check-in/check-out processes
- Improved coordination between hotel departments
- Quick access to reports and operational data
- Enhanced data security and record management
- Reduced manual paperwork and administrative effort

3.3 Features of Proposed System

- Manage room details and availability status
- Create, update, and cancel reservations
- Register and maintain guest information
- Process guest check-in and check-out activities
- Generate bills and track payment transactions
- Manage staff records and responsibilities

This system brings all essential hotel management functionalities into a single, easy-to-use software application, improving operational efficiency, customer satisfaction, and overall business management.

4. Hardware Requirements

The hardware requirements for this project are moderate because the Hotel Management System is designed to handle room reservations, guest records, billing operations, staff management, and report generation. A stable internet connection may be required for database synchronization, online reservations,

4.1 Minimum Requirements

Component	Requirement
Processor	Intel / AMD Dual Core
RAM	4 GB or above
Storage	1 GB free space
Monitor	Any basic display
Keyboard	Required for input
Internet	Required for online booking and data synchronization
Operating System	Windows / macOS / Linux

4.2 Recommended Requirements

Component	Recommended
Processor	Intel i5 or higher
RAM	8 GB or more
Storage	5 GB
Internet	Broadband — stable connection
OS	Windows 10 / Ubuntu 20.04 or above

Since the system manages multiple hotel operations such as reservations, guest records, billing, and reporting, mid-range hardware is sufficient for smooth performance. For hotels with a large number of users and transactions, higher hardware specifications are recommended to ensure optimal efficiency and responsiveness.

5 . Software Requirements

5.1 Software Used

Software	Purpose
Python 3.8+	Core programming language
MySQL	Database management and data storage
VS Code / Eclipse / IntelliJ IDEA	Code editor / IDE
MySQL Workbench	Database design and administration
Console / Terminal	User input and output interface

5.2 Required Software Versions

- Python 3.8 or above / Java JDK 8 or above
- MySQL 8.0 or above
- MySQL Connector (latest stable version)
- VS Code, Eclipse, or IntelliJ IDEA
- Windows 10 / Linux operating system

5.3 Additional Tools (Optional)

- Git — for version control
- XAMPP / WAMP — for local server hosting
- Microsoft Excel — for report export and analysis
- Word Processor — for documentation
- Draw.io / StarUML — for UML diagram creation

6.Module Description

The entire Hotel Management System is divided into eight logical modules. Each module has a clearly defined responsibility that contributes to the overall system functionality.

Module	Class / File	Primary Role
User Management Module	User Manager	Manages user accounts and access permissions
Room Management Module	Room Manager	Handles room details and availability
Reservation Module	Reservation Manager	Manages room booking and cancellation
Guest Management Module	Guest Manager	Stores and maintains guest information
Check-In / Check-Out Module	Check In Manager	Processes guest arrivals and departures
Billing Module	Billing Manager	Generates invoices and manages payments
Reporting Module	Report Generator	Creates operational and financial reports
User Interface Module	ConsoleUI	Manages menus and user interaction

6.1 User Management Module

This module manages user authentication and authorization within the system. It ensures secure access to hotel operations.

Responsibilities:

- Create and manage user accounts
- Authenticate users during login
- Assign roles and access permissions
- Maintain user profile information

6.2 Room Management Module

This module manages room information and availability status.

Responsibilities:

- Add, update, and delete room records
- Track room availability in real time
- Update room status (Available, Occupied, Maintenance)
- Provide room details for reservations

6.3 Reservation Module

This module handles the room booking process for guests.

Responsibilities:

- Create new room reservations
- Modify existing bookings
- Cancel reservations when required
- Check room availability before booking

6.4 Guest Management Module

This module stores and manages guest-related information.

Responsibilities:

- Register new guests
- Maintain guest personal details
- Store guest stay history
- Retrieve guest records when required

6.5 Check-In / Check-Out Module

This module manages guest arrival and departure operations.

Responsibilities:

- Process guest check-in requests
- Assign rooms to guests
- Handle guest check-out procedures
- Update room occupancy status automatically

6.6 Billing Module

This module manages hotel billing and payment processing.

Responsibilities:

- Generate invoices for guest stays
- Calculate room charges and additional services
- Record payment transactions
- Maintain billing history

6.7 Reporting Module

This module generates various reports required by hotel management.

Responsibilities:

- Generate occupancy reports
- Generate revenue and billing reports
- Produce guest activity reports
- Provide management insights for decision-making

6.8 User Interface Module

This module serves as the interaction layer between users and the system.

Responsibilities:

- Display menus and navigation options
- Accept user input
- Display booking, billing, and guest information
- Provide a user-friendly and responsive interface

7. Functional Requirements

Functional requirements describe what the system must do. Each requirement is mapped to a specific feature of the Hotel Management System.

FR No.	Requirement	Description
FR1	User Login	Authorized users can securely log in to the system using valid credentials
FR2	Room Management	System allows adding, updating, deleting, and viewing room details
FR3	Room Reservation	System enables guests' room booking, modification, and cancellation
FR4	Guest Registration	System stores and manages guest information and records
FR5	Check-In Processing	System allows hotel staff to check guests into assigned rooms
FR6	Check-Out Processing	System processes guest check-out and updates room availability
FR7	Billing	System automatically generates bills and records payment transactions
FR8	Staff Management	System manages employee details, roles, and responsibilities
FR9	Report Generation	System generates occupancy, revenue, and guest activity reports
FR10	Logout / Exit	User can safely log out and exit the application at any point

8. Non-Functional Requirements

Non-functional requirements describe the quality attributes, performance characteristics, and constraints of the system.

NFR No.	Attribute	Description
NFR1	Usability	The system interface must be simple, user-friendly, and easy for hotel staff to operate with minimal training
NFR2	Performance	Room booking, guest management, billing, and report generation operations should execute efficiently with minimal response time
NFR3	Reliability	The system must provide accurate data processing and robust error handling to ensure uninterrupted hotel operations
NFR4	Scalability	The system should support increasing numbers of rooms, guests, staff, and transactions without significant performance degradation
NFR5	Portability	The system should run on Windows and Linux platforms with the required software installed
NFR6	Maintainability	The application should be modular, well-documented, and easy to update or enhance in the future
NFR7	Security	User authentication, role-based access control, and secure storage of sensitive hotel and guest information must be implemented
NFR8	Availability	The system should be available whenever required and ensure continuous access to hotel management functions during business operations

9. High-Level Design (HLD)

9.1 Overall System Architecture / Flow

The following diagram illustrates the high-level operation of the Hotel Management System. It shows the four main processing layers — Input Layer, Management Processing Layer, Database Layer, and Output Layer — and how data flows between each component from user requests through to report generation and hotel operations management.

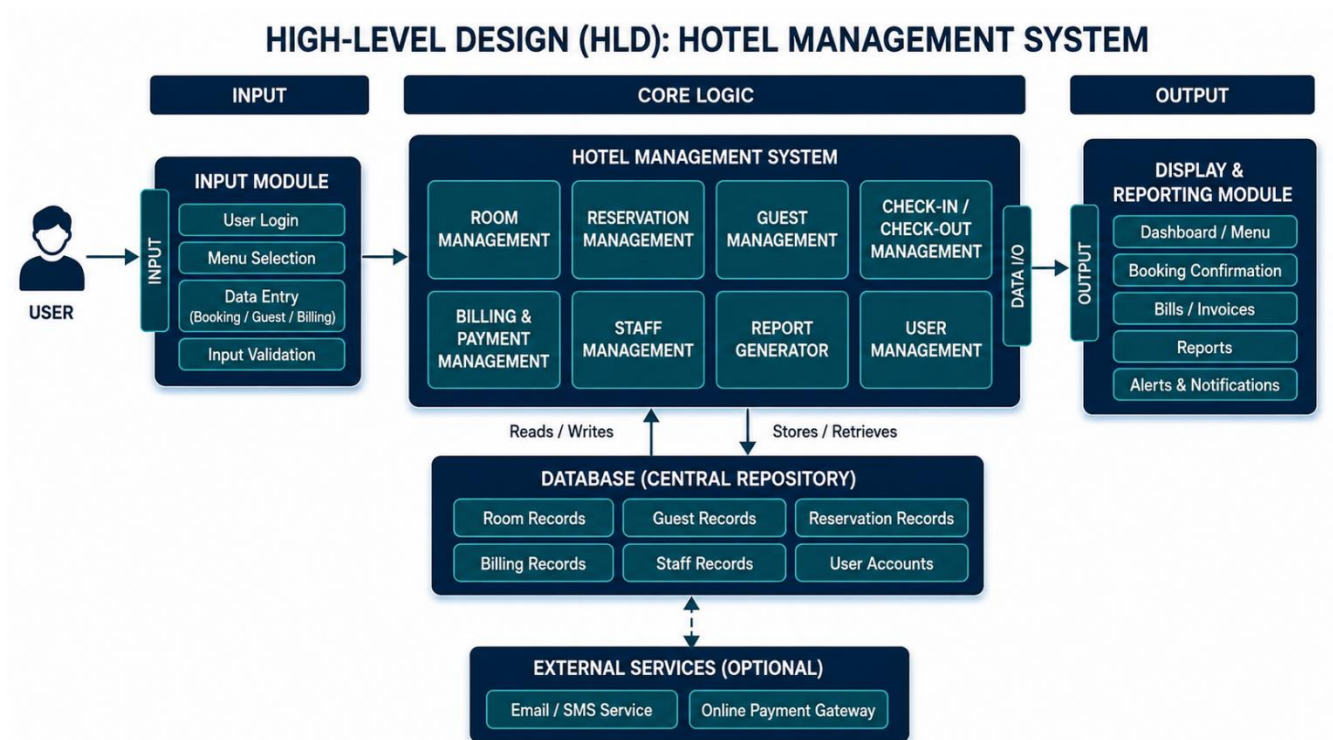


Figure 9.1 — High-Level Design Architecture Diagram

10. Low-Level Design (LLD)

The Low-Level Design describes the internal structure of each module and the detailed interactions between classes. It specifies the attributes and methods of each class and shows how they collaborate to deliver the system's functionality.

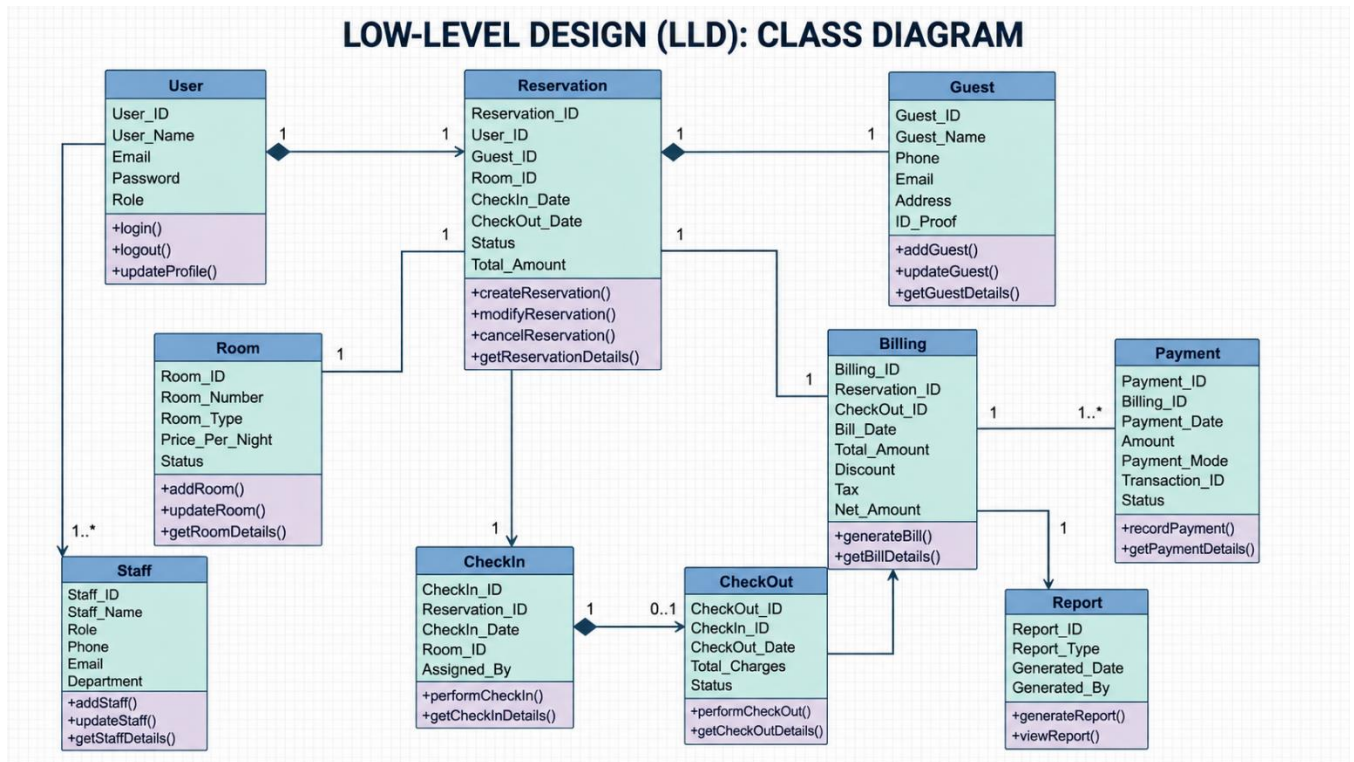


Figure 10.1 — Low-Level Design Class Structure Diagram

11.UML Diagrams

UML (Unified Modeling Language) is a standard visual modeling language used to design and document software systems. The following UML diagrams represent the structure, behaviour, and interactions of the Hotel Management System.

Diagram	Purpose
Use Case Diagram	Shows actors and their interactions with system use cases
Class Diagram	Illustrates the object-oriented class structure and relationships
Sequence Diagram	Captures the chronological message flow between components
Activity Diagram	Shows the step-by-step flow of execution from start to end
ER Diagram	Shows data entities, attributes, keys, and their relationships

11.1 Use Case Diagram

The Use Case Diagram identifies the **Customer**, **Receptionist**, and **Housekeeper** as the primary actors and shows their interactions with the core functionalities of the Hotel Management System.

The diagram illustrates how customers can check room availability, book rooms, cancel reservations, make payments, and check out, while receptionists manage reservations and guest services. The housekeeper is responsible for providing room services and maintaining room cleanliness. Relationships between use cases are represented using the «include» stereotype for dependent functionalities such as refunds, meal services, and room cleaning.

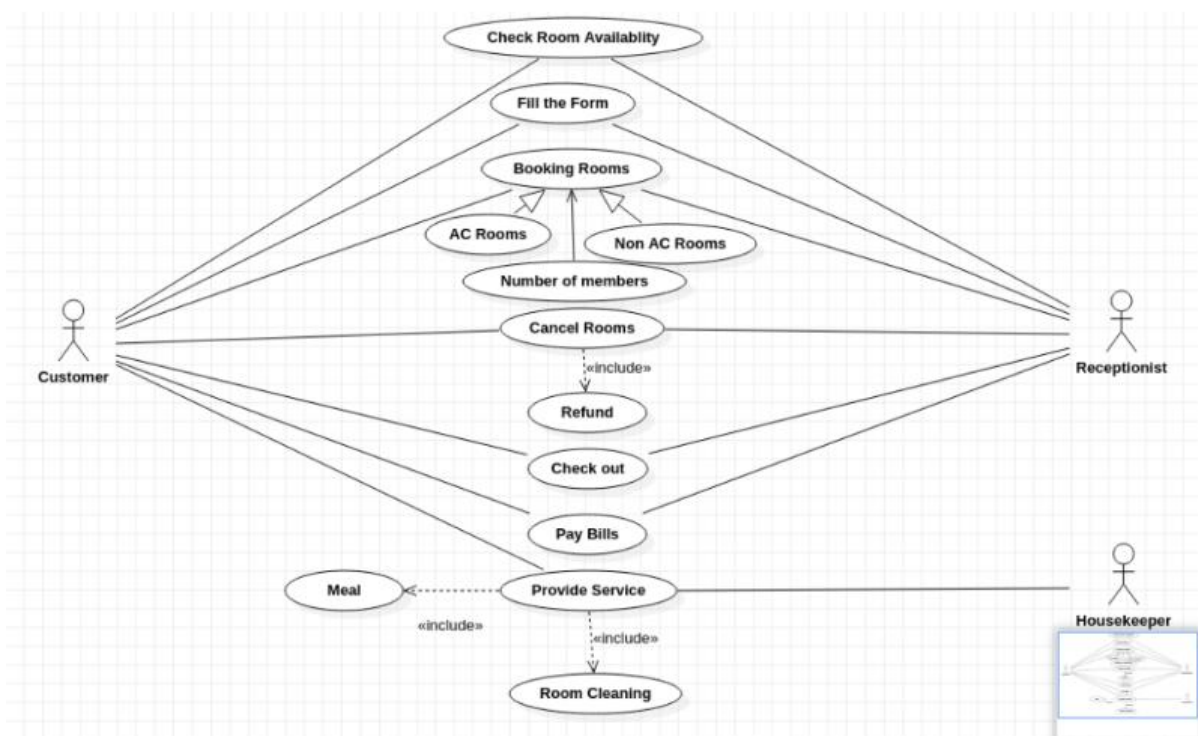


Figure 11.1 — Use Case Diagram

11.2 Class Diagram

The Class Diagram illustrates the object-oriented structure of the Hotel Management System. Nine classes are defined — User (manages authentication and system access), Room (stores room details and availability), Reservation (handles room bookings), Guest (maintains guest information), CheckIn (manages guest arrivals), CheckOut (handles guest departures), Billing (generates invoices and charges), Payment (records payment transactions), and Report (generates operational and financial reports).

Relationships include associations, aggregation/composition, and navigability between classes. The diagram represents the complete flow of hotel operations from reservation and guest management to billing, payment processing, and report generation..

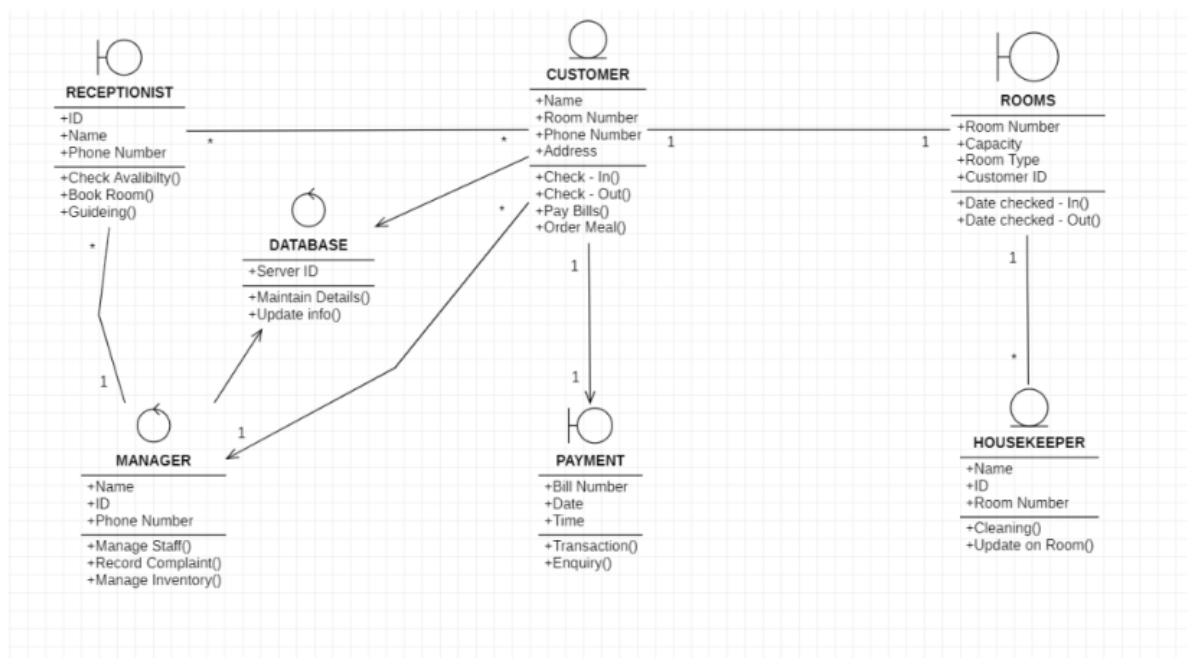


Figure 11.2 — Class Diagram

11.3 Sequence Diagram

The Sequence Diagram captures the chronological message flow between the Customer, Receptionist, Hotel Management System, and Database across three phases: Reservation Phase (room availability checked, booking details submitted, reservation confirmed), Guest Service Phase (check-in, room allocation, and service requests processed), and Billing Phase (bill generation, payment processing, and check-out completion).

The diagram illustrates how information is exchanged between users and the system to ensure smooth hotel operations and efficient guest management.

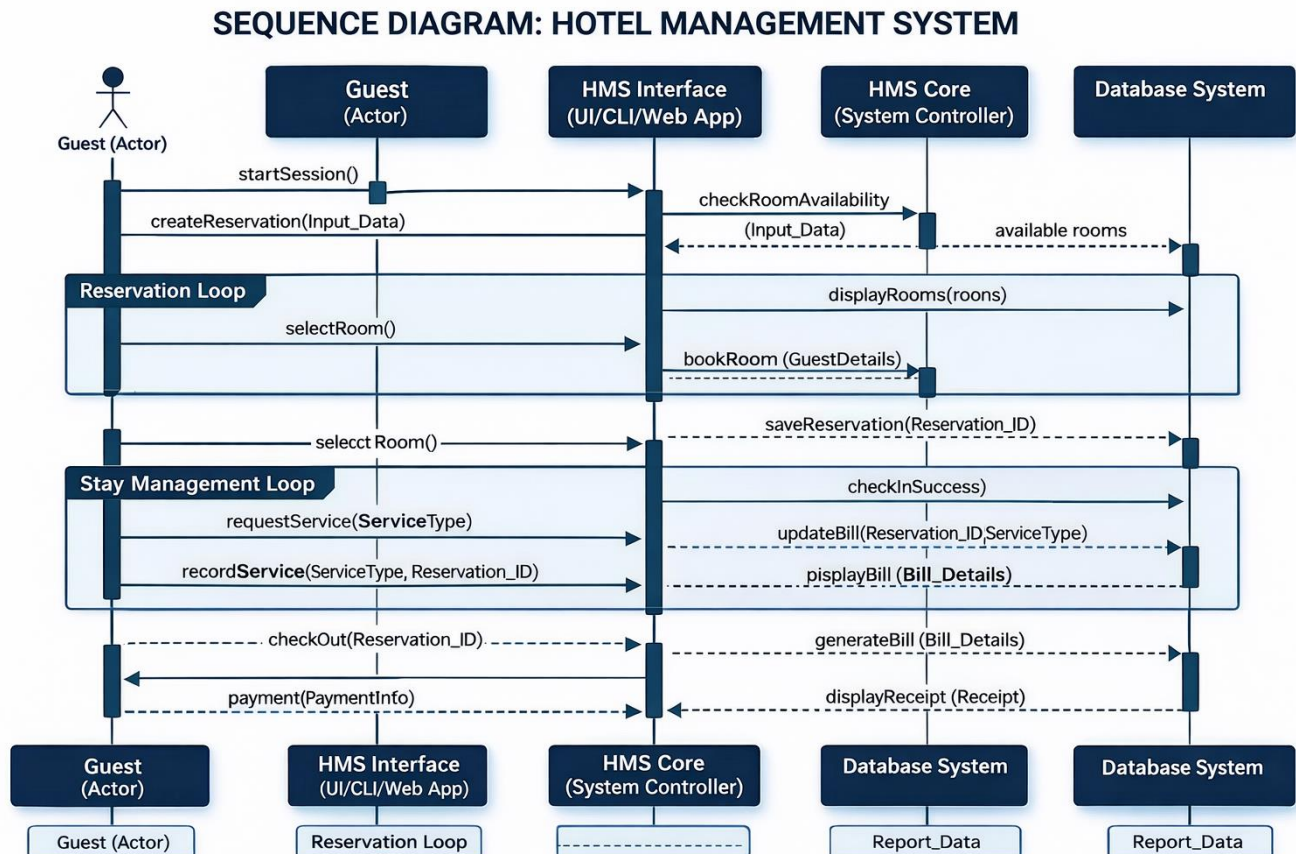


Figure 11.3 — Sequence Diagram

11.4 Activity Diagram

The Activity Diagram illustrates the overall workflow of the Hotel Management System. It begins with user login, followed by room reservation and availability checking. Once a room is confirmed, the guest checks in and can access hotel services during their stay. The system records services, updates billing details, and processes check-out. Finally, the bill is generated, payment is completed, reports are created, and the session ends. Decision diamonds are used to verify room availability and determine whether additional services are required before check-out.

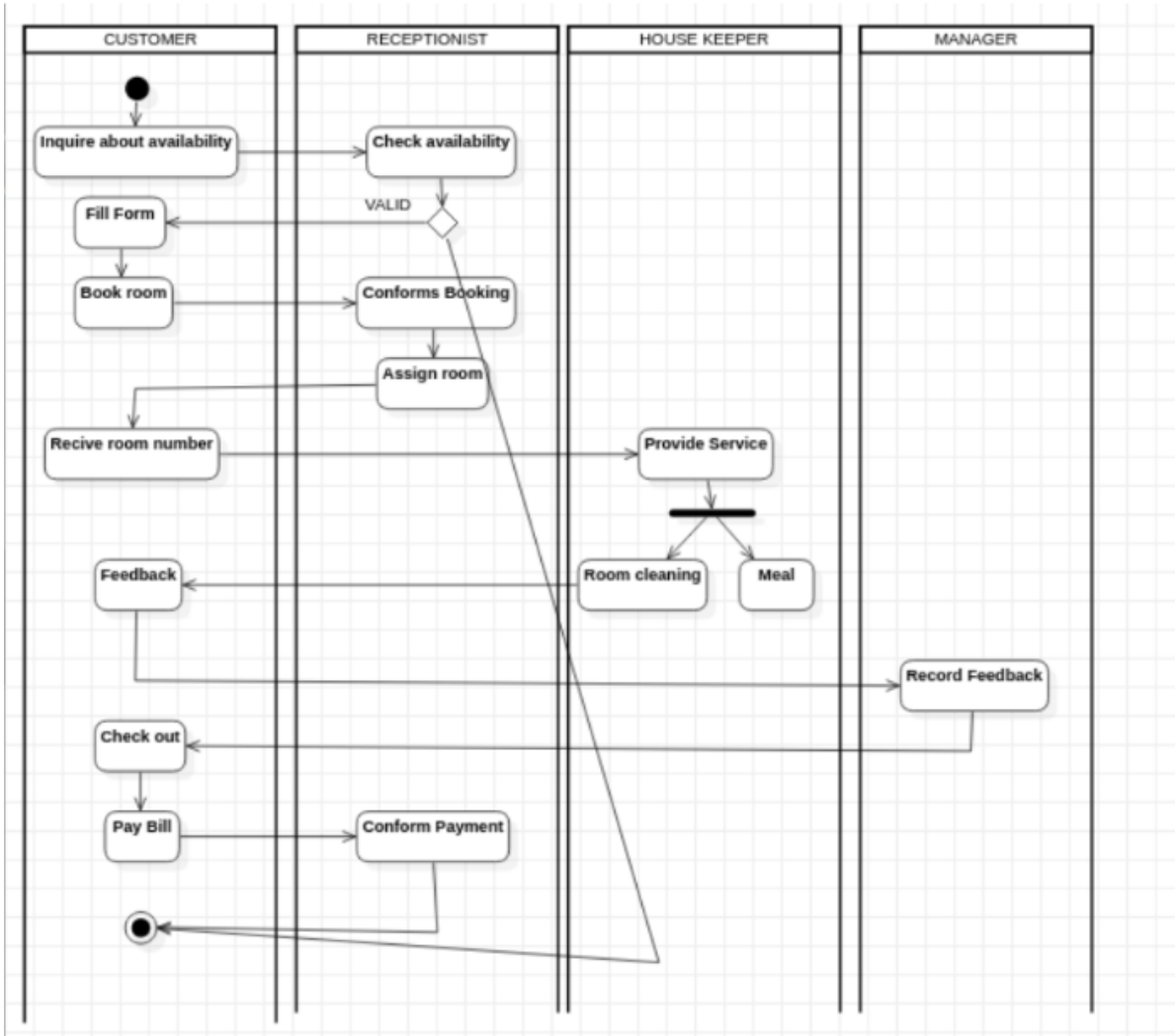


Figure 11.4 — Activity Diagram

11.5 ER Diagram

The Entity-Relationship (ER) Diagram represents the six key data entities in the Hotel Management System **GUEST, RESERVATION, ROOM, SERVICE, BILL, and PAYMENT** along with their attributes, primary keys (PK), foreign keys (FK), and relationships with cardinality notation (1:1 and 1:N). The diagram illustrates how guests make reservations, rooms are allocated, services are utilized, bills are generated, and payments are processed. It defines the logical data model that supports hotel operations such as reservation management, guest services, billing, and payment processing.

ER DIAGRAM: HOTEL MANAGEMENT SYSTEM

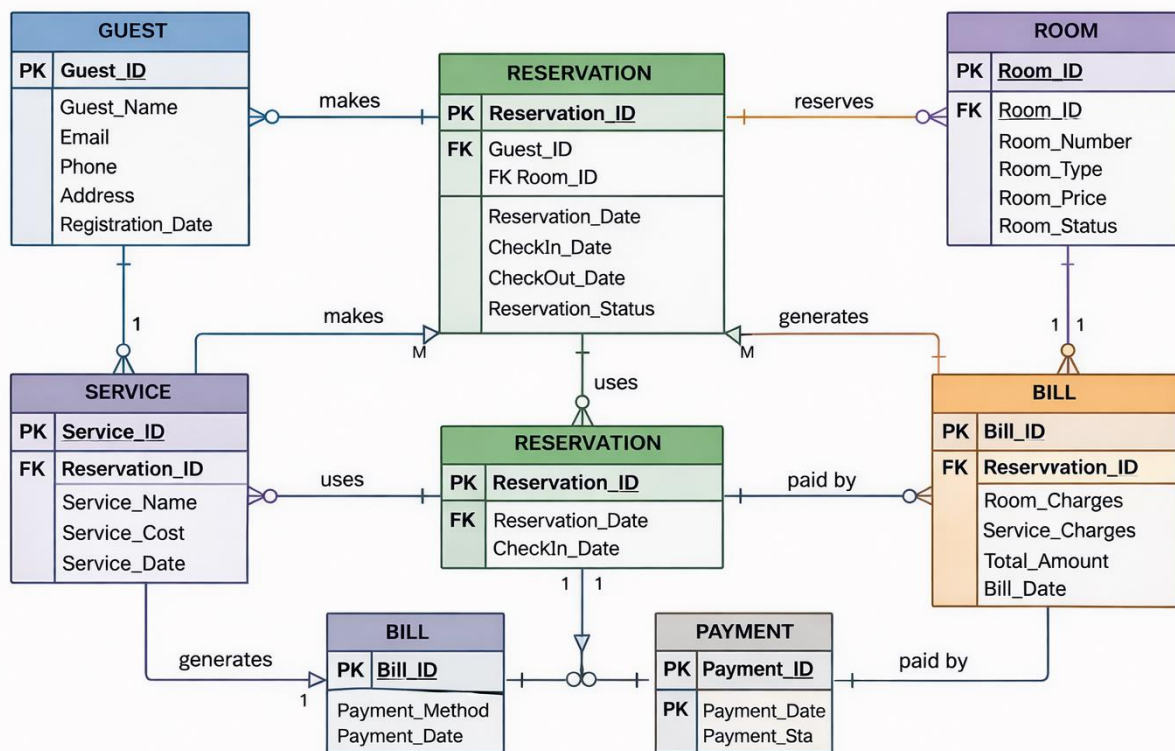


Figure 11.5 — Entity-Relationship (ER) Diagram

12 . Proposed Test Cases

The following test cases are designed based on the functional requirements of the Hotel Management System. These represent planned validation scenarios covering room reservation, guest management, check-in/check-out, billing, payment processing, and error handling.

12.1 Test Case Table

TC ID	Test Description	Input	Expected Output	Type	Status
TC01	Valid Room Reservation	Guest details and available room selection	Reservation created successfully	System	Planned
TC02	Invalid Room Reservation	Missing guest details or invalid dates	System displays validation error message	System	Planned
TC03	Room Availability Check	Request for available room	Available rooms displayed correctly	System	Planned
TC04	No Rooms Available	Reservation request when all rooms are occupied	System displays "No Rooms Available" message	Unit	Planned
TC05	Guest Check-In	Valid reservation ID	Guest check-in completed successfully	System	Planned
TC06	Invalid Check-In	Invalid or non-existent reservation ID	System displays appropriate error message	Unit	Planned
TC07	Guest Check-Out	Valid checked-in guest	Final bill generated and room status updated	Integration	Planned
TC08	Service Addition	Room service request during stay	Service recorded and bill updated	Integration	Planned
TC09	Payment Processing	Valid payment details	Payment successful and receipt generated	System	Planned
TC10	Payment Failure Handling	Invalid payment information	Error message displayed and transaction cancelled	System	Planned

12.2 Test Case Note

Note: The above test cases represent expected system behaviour based on defined functional requirements. These are planned validation scenarios prepared at the design stage of the project. Two additional test cases (TC09 and TC10) have been included to improve coverage of edge cases

13. Future Scope

The current version of the Hotel Management System is designed to automate hotel operations such as room reservations, guest management, billing, and report generation. The system can be extended in the following ways to improve its scalability, usability, and real-world applicability:

Enhancement	Description
Web-Based Application Interface	Upgrade the system to a fully interactive web application using Flask, Django, or ASP.NET with a responsive user interface.
Online Room Booking Portal	Allow guests to book rooms online through a dedicated website with real-time room availability.
Database Integration	Integrate MySQL, PostgreSQL, or SQLite to securely store guest details, reservations, billing records, and reports.
User Authentication & Role Management	Implement secure login for administrators, receptionists, and hotel staff with role-based access control.
Online Payment Gateway	Integrate payment gateways such as Razorpay, PayPal, or Stripe for secure online payments and booking confirmations.
Automated Email & SMS Notifications	Send booking confirmations, check-in reminders, payment receipts, and promotional messages automatically.
Advanced Reporting & Analytics	Generate detailed reports on occupancy rates, revenue, customer trends, and hotel performance.
Mobile Application Version	Develop Android and iOS applications for guests and hotel staff to access services on mobile devices.
Multi-Hotel Management Support	Extend the system to manage multiple hotel branches from a centralized platform.
AI-Based Customer Support	Integrate AI chatbots to handle guest inquiries, room recommendations, and booking assistance.

These enhancements will significantly improve the scalability, usability, and real-world applicability of the system.

14. Conclusion

The Hotel Management System is an integrated software solution developed to automate and streamline various hotel operations, including room reservations, guest management, check-in and check-out processes, billing, staff management, and report generation. The system provides an efficient and centralized platform for managing hotel activities, improving operational productivity and customer service.

The application manages guest records, room availability, reservations, billing transactions, and hotel services automatically, reducing manual effort and minimizing operational errors. It enables hotel staff to perform daily tasks efficiently while providing management with accurate information and real-time access to business data.

The system follows a structured modular architecture and clearly demonstrates the practical application of software engineering principles such as object-oriented design, database management, modular programming, and user interface development. It showcases how technology can be effectively utilized to improve the efficiency and reliability of hotel operations.

This project serves as a strong foundational model for future enhancements such as online booking integration, mobile application support, cloud-based deployment, and advanced analytics dashboards, reflecting the growing importance of digital solutions in the hospitality industry.

